METRIC

MIL-PRF-8565/14(AS)
w/AMENDMENT 1
24 March 2008
SUPERSEDING
MIL-PRF-8565/14(AS)
25 June 2001

PERFORMANCE SPECIFICATION SHEET

BATTERY, STORAGE, AIRCRAFT, MEDIUM RATE, MAINTENANCE-FREE, 24-VOLT, 15-AMPERE-HOUR

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-8565.

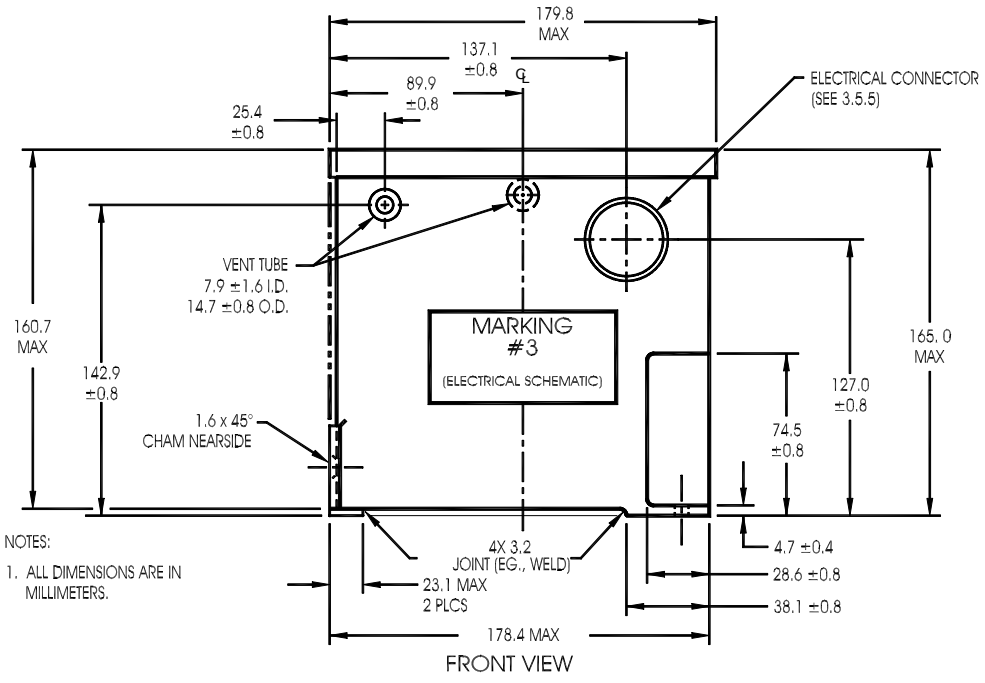


FIGURE 1. Dimensions and configuration for PIN D8565/14-1.

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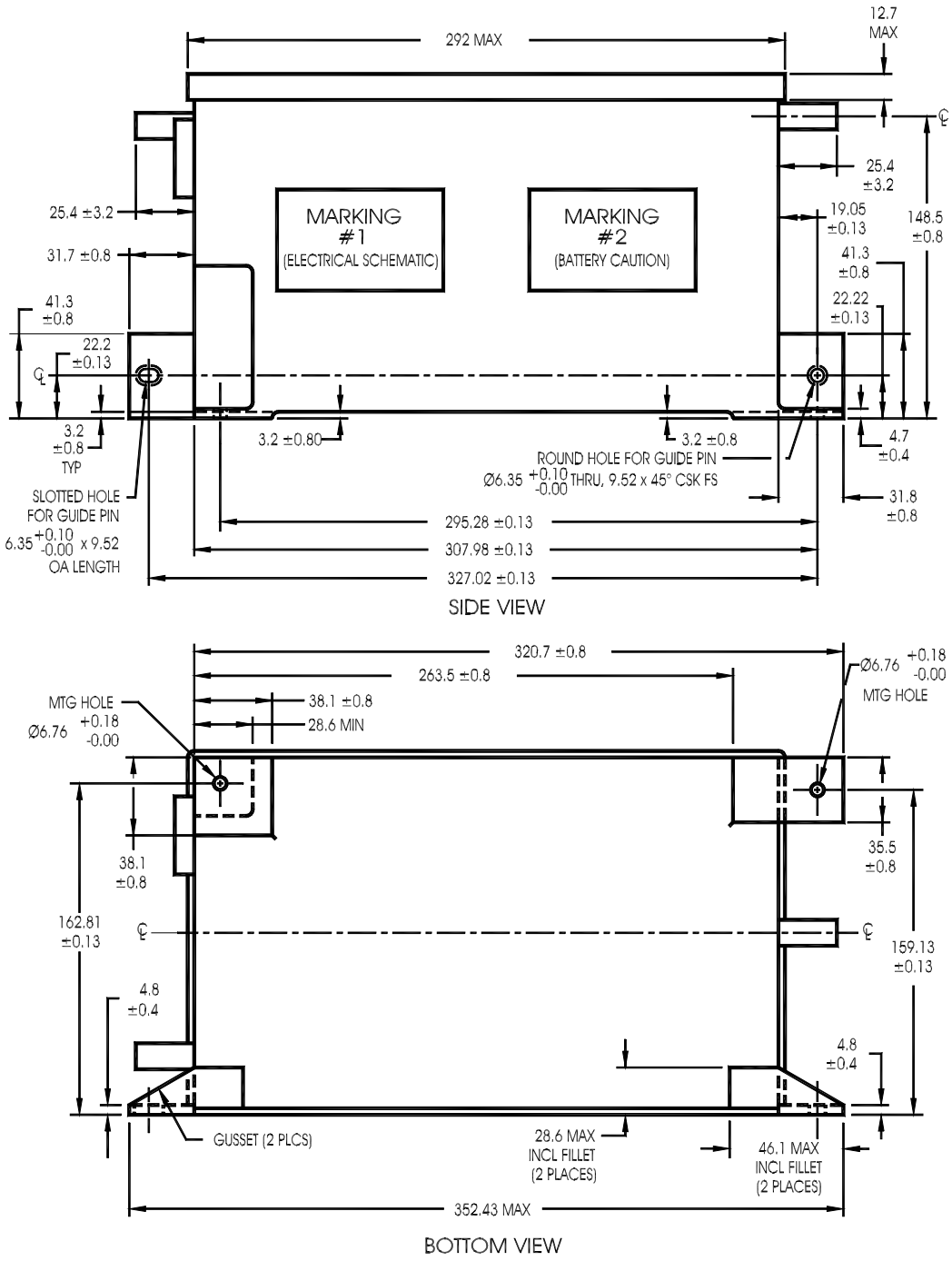


FIGURE 1. Dimensions and configuration for PIN D8565/14-1 - Continued.

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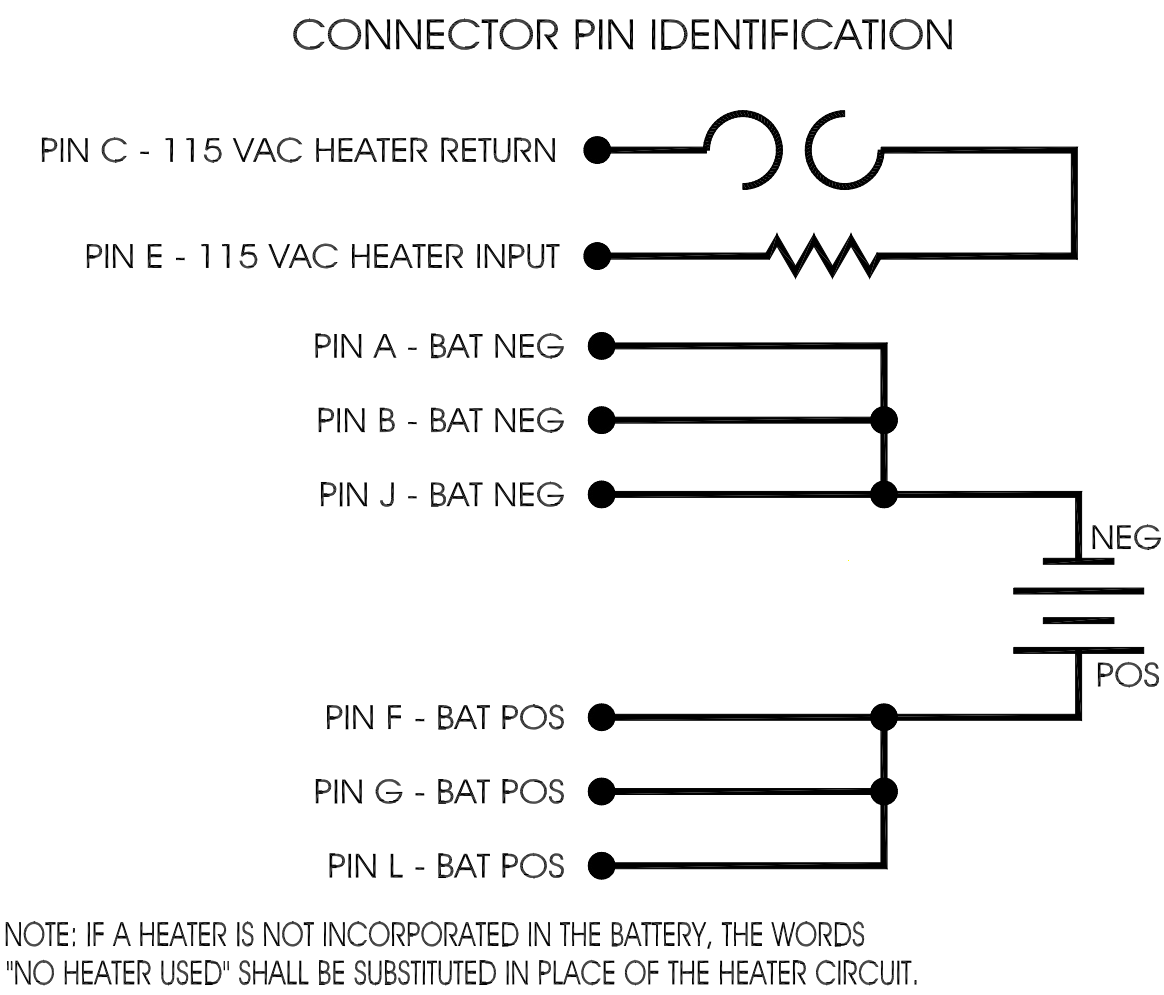


FIGURE 2. Electrical schematic for marking #3.

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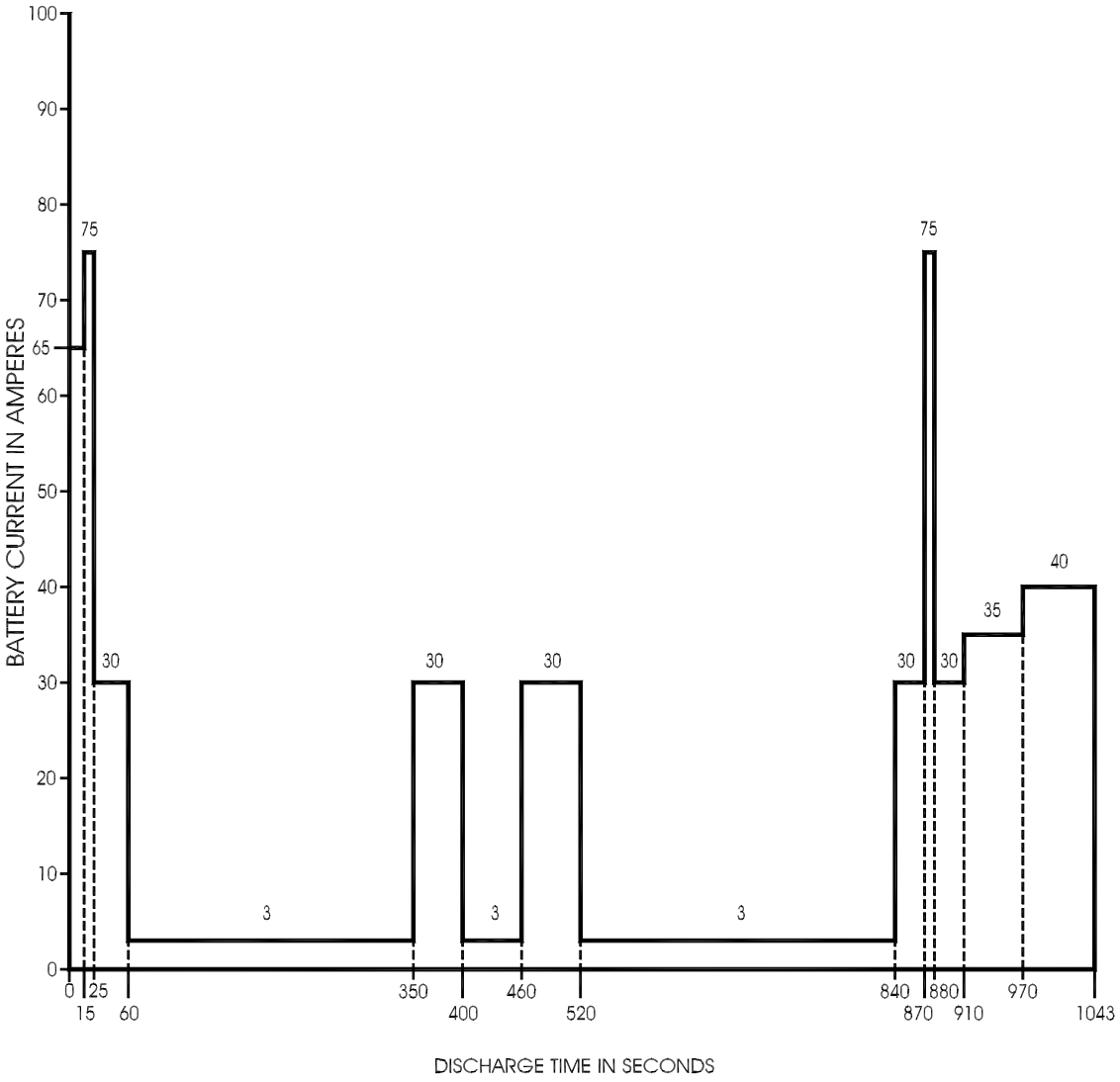


FIGURE 3. Aircraft battery discharge profile requirements.

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FREQUENCY (Hz)	PERFORMANCE AMPLITUDE (g^2/Hz) OR SLOPE	ENDURANCE AMPLITUDE (g^2/Hz) OR SLOPE
10-1000 1000-2000	0.04 -6dB/OCTAVE	0.04 -6dB/OCTAVE

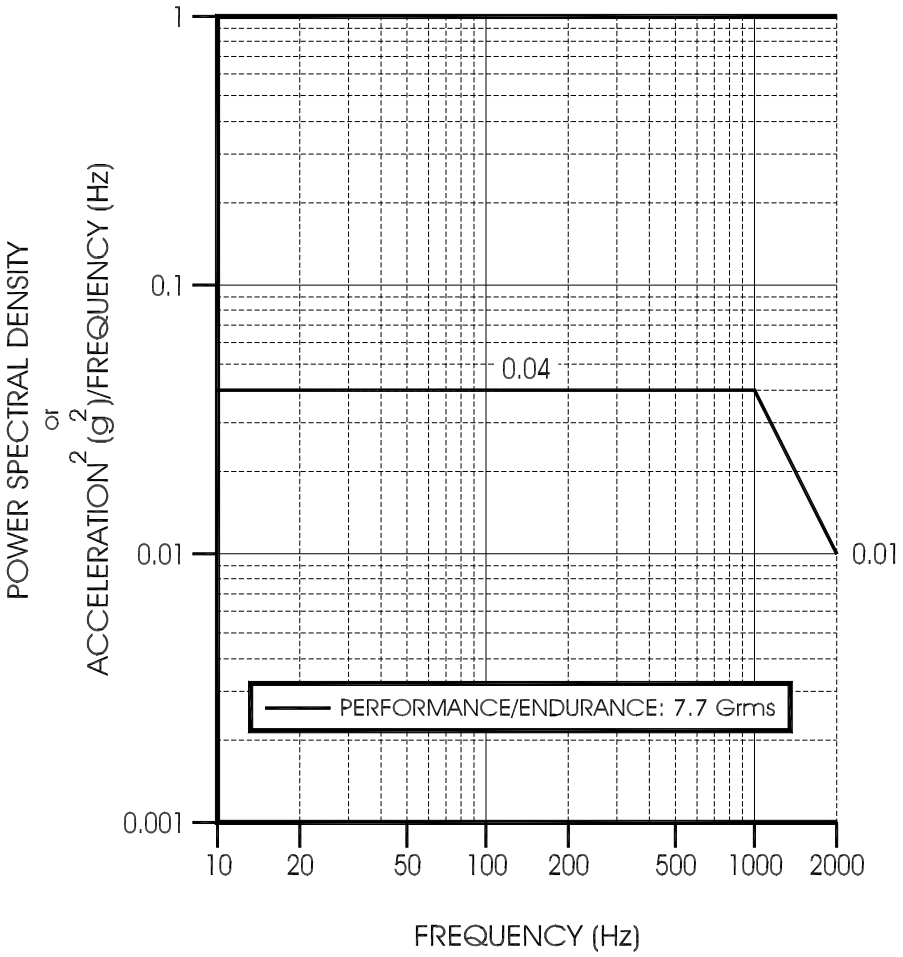


FIGURE 4. Random vibration test levels.

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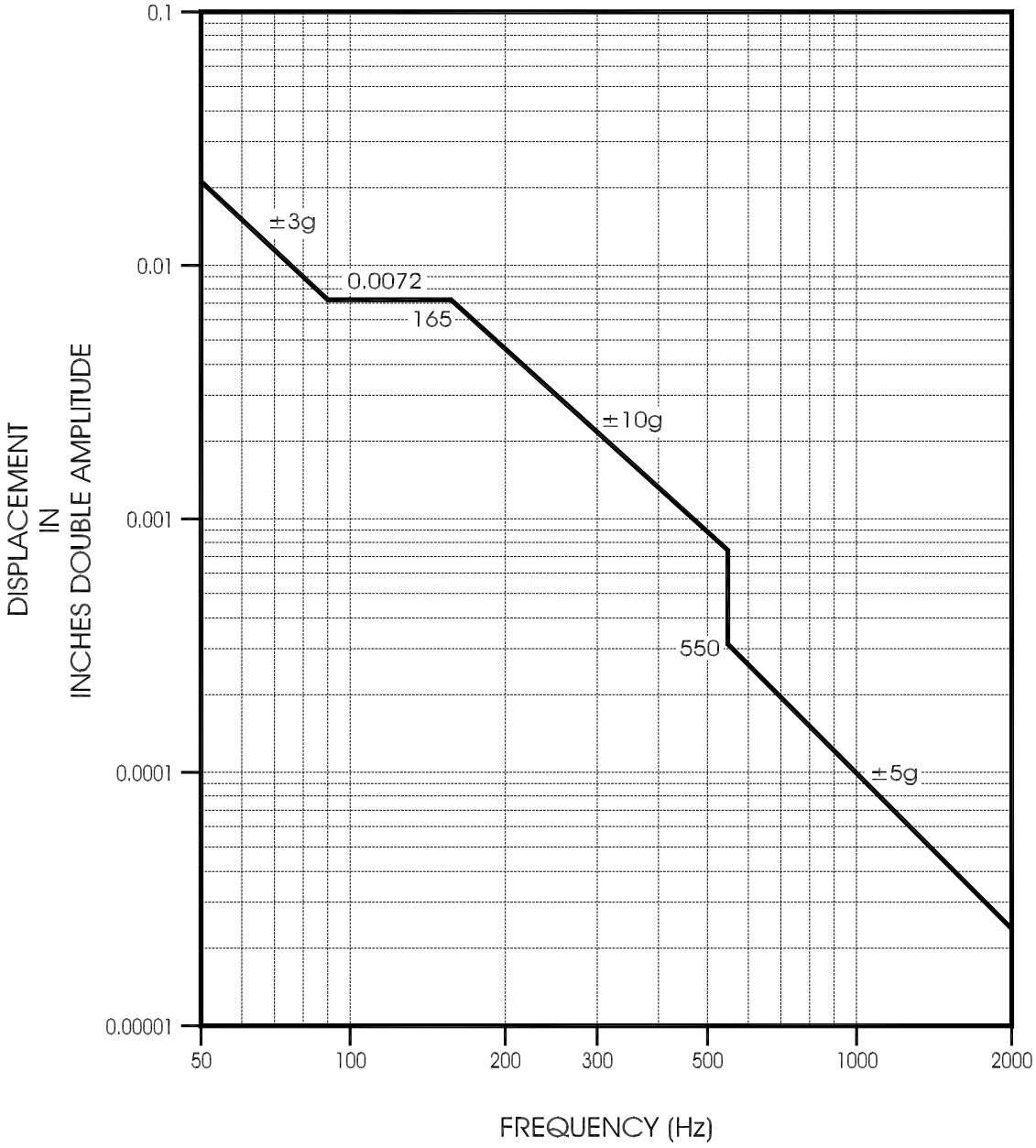


FIGURE 5. Sinusoidal vibration test levels.

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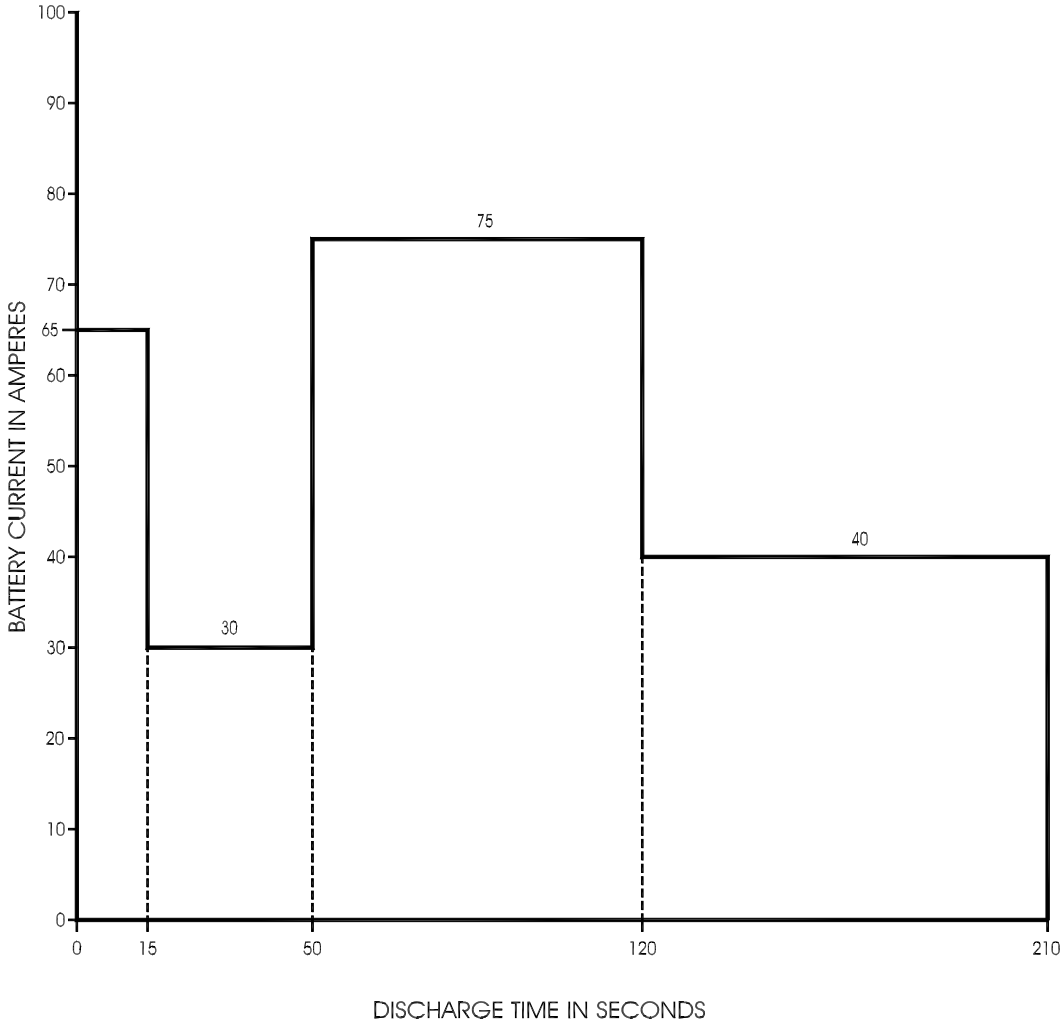


FIGURE 6. Aircraft start-up load requirements.

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REQUIREMENTS:

1. Part or identifying number. The part or identifying number (PIN) of the battery covered by this specification sheet shall be D8565/14-1.

2. Dimensions and weight. The dimensions of the battery shall be as shown on figure 1 contained in this specification sheet. The weight of the battery shall be not greater than 18.7 kilograms (see 3.8.3 and 4.5.3).

3. Capacity. The rated capacity of the battery shall be 15.0 ampere-hours (Ah). See 3.8.6 as modified herein.

4. Recharge characteristics. The battery, after being discharged at the 1C-rate to 20.0 volts, shall successfully recharge to its full capacity in 2.0 ± 0.1 hours when constant potential charged at 28.25 ± 0.25 volts with charge currents of 50.0 ± 1.0 amperes at 24°C (75°F) and 43°C (109°F) (see 4.5.10.2 and 6.11.7.1, as modified herein).

5. Design characteristics. The battery shall use a starved-electrolyte, sealed electrochemical technology with pressure-regulated, self-resealing safety valve(s). Electrolyte leakage or spillage from the battery shall not occur during aerobatic maneuvers of the aircraft.

6. Service life. The battery's service life (useful life) in aircraft shall be not less than 2 years. The battery shall require no scheduled maintenance during the 2 years (see 3.8.9, 4.5.10, and 6.11.7.3).

7. Qualification inspection and conformance inspection. Table I lists the required inspections.

8. MIL-PRF-8565 variance. The battery shall comply with MIL-PRF-8565 except as follows.

8.1 The following paragraphs and figures are not applicable:
- 3.8.7 Constant voltage discharge.
- 3.8.8 Strength of receptacle.
- 4.5.8 Constant voltage discharge.
- 4.5.9 Strength of receptacle test.
- 4.5.11 Capacity discharge at 49°C (120°F).
- 4.5.12 Charge and discharge test at low temperature.
- 4.5.30 Wormgear retaining pin hardness.
- FIGURE 3. Battery mounting.

8.2 Modify the following paragraphs and table to delete "18.0 volts" and insert "20.0 volts".
- 4.5.6 Electrical test preparation, first sentence.
- 4.5.7 Capacity discharge, second sentence.
- 4.5.17 Temperature rise and float test, step 4.5.17.e, second sentence.
- 4.5.22 Ground storage test, step 4.5.22.d.
- TABLE III. Capacity performance requirements, requirement number (1).

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8.3 Modify the following paragraphs:

3.5.5 Receptacles. Insert the following after the first sentence: "The battery's electrical interface to the aircraft shall be in accordance with MIL-C-38999, series III and equivalent to any of the following receptacles: PIN D38999/20KG11SN, PIN D38999/24KG11SN, PIN D38999/20YG11SN, or PIN D38999/24YG11SN. It shall be positioned and electrically connected as shown on figures 1 and 2, respectively, of the specification sheet."

3.5.10 Heaters. In 3.5.10.b, delete the text and insert the following: "Heater current shall be not greater than 3.0 amperes. The rated current of each heater control device shall be compatible with the heater current."

8.4 Add the following paragraph:

"3.5.11 Mounting holes. The mounting surfaces that surround each of the two front mounting holes shall be free of any coating for a distance not less than 12.7 millimeters radius from the center of the hole. This shall apply to the top of the mounting surface and is recommended for the bottom surface."

8.5 Modify the following paragraphs:

3.6.5 Marking for batteries with heaters. At the end of the third sentence, insert: "(see figure 2 of the specification sheet)."

3.8.6 Capacity and electrical performance. Delete the entire text and insert: "The rated capacity of the battery shall be not less than 15 ampere-hours. To meet the capacity requirement, each battery shall produce a potential of not less than 20.0 volts at not less than 60 minutes when connected to a 15-ampere load after being tested in accordance with 4.5.7 at 24°C (75°F) (15.0 Ah/1.0 hr/24°C/20.0 V at end of life). The battery's 1C-rate shall be 15.0 amperes (see requirement 3 of the specification sheet)."

3.8.9 Cycling. Delete the text.

8.6 Add the following paragraphs:

"3.8.9.1 Qualification inspection. The battery terminal discharge voltage shall be not less than 18.0 volts when performing each of the 400 discharge-charge-rest cycles of 4.5.10.1. In addition, the battery shall deliver not less than 10 ampere-hours when discharged in accordance with step 4.5.10.1.e and not less than 15 ampere-hours when discharged in accordance with step 4.5.10.1.g. The battery shall also comply with the service life requirement specified in requirement 6 of the specification sheet.

"3.8.9.2 Conformance inspection. The battery terminal discharge voltage shall be not less than 20.0 volts when performing each of the 100 discharge-charge-rest cycles of 4.5.10.2. In addition, the inability of the battery to meet requirement (1) of table III for three consecutive

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discharges shall constitute a failure. The battery shall also comply with requirement 4 of the specification sheet."

8.7 Modify the following paragraphs:

3.8.16 Output performance. Delete the text and insert: "The battery shall produce a potential of not less than 18.0 volts during and after the tests of 4.5.28."

3.8.17 Cold and hot temperature charge. Delete the title and first sentence and insert: "Ultra-cold temperature charge. The battery shall produce not less than 13.5 ampere-hours after being tested in accordance with 4.5.29."

8.8 Add the following paragraphs:

"3.8.18 Vibration. The battery shall produce a potential of not less than 20.0 volts after each of the discharges of steps 4.5.19.c, 4.5.19.d, and 4.5.19.h. The battery shall also meet the requirements of 3.7 and 3.8.10 after being tested in accordance with 4.5.19.

"3.8.19 Hot-temperature charge. The battery shall perform as follows after being tested in accordance with 4.5.32.

"a. Sample 1 (stabilized at 50°C): This sample shall provide not less than 15.0 ampere-hours of capacity after each of the discharges of steps 4.5.32.b and 4.5.32.f. The terminal voltage of the battery shall be not less than 18.0 volts after being discharged in accordance with step 4.5.32.d. The battery shall also meet the requirements of 3.7.

"b. Sample 2 (stabilized at 68°C): This sample shall provide not less than 13.5 ampere-hours of capacity after each of the discharges of steps 4.5.32.b and 4.5.32.f. The terminal voltage of the battery shall be not less than 18.0 volts after being discharged in accordance with step 4.5.32.d. The battery shall also meet the requirements of 3.7."

8.9 Modify the following paragraphs:

4.4.2 Constant potential charging method. In the second sentence, delete "not less than 8 times the capacity rating of the battery being tested" and insert "50.0 ± 1.0 amperes".

4.5.10 Cycling test. Delete the text.

8.10 Add the following paragraphs and table:

"4.5.10.1 Qualification testing. Simultaneously subject the battery to 400 cycles of discharge-charge-open circuit (rest) and to the temperature environments specified in table II of the specification sheet. This test consists of the following steps.

"a. Constant potential charge the battery in accordance with 4.4.2.

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"b. First sequence (cycles 1 through 75).

"(1) Room temperature environment:

"(a) Stabilize the battery temperature at 24°C (75°F). Perform steps 4.5.10.1.b(1)(b) through 4.5.10.1.b(1)(e) with the battery in the 24°C (75°F) temperature environment.

"(b) Discharge the battery in accordance with figure 3 of the specification sheet; however, the discharge time at 40 amperes shall be increased from 73 seconds to 110 seconds, which increases the total discharge time from 1043 seconds to 1080 seconds.

"(c) Immediately after completing step 4.5.10.1.b(1)(b), constant potential charge the battery in accordance with 4.4.2.

"(d) Open circuit the battery terminals for a period of 102.0 ± 1.0 minutes.

"(e) Repeat above steps 4.5.10.1.b(1)(b) through 4.5.10.1.b(1)(d) for a total of 25 cycles.

"(2) Hot temperature environment: Repeat step 4.5.10.1.b(1), except the specified temperature shall be changed from 24°C (75 °F) to 43°C (109 °F).

"(3) Cold temperature environment: Repeat step 4.5.10.1.b(1), except the specified temperature shall be changed from 24°C (75°F) to –18°C (-0.4°F).

"c. Repeat steps 4.5.10.1.b(1) through 4.5.10.1.b(3) for each of the following sequences.

"(1) Second sequence (cycles 76 through 150).

"(2) Third sequence (cycles 151 through 225).

"(3) Fourth sequence (cycles 226 through 300).

"(4) Fifth sequence (cycles 301 through 375).

"d. Repeat step 4.5.10.1.b(1) for the sixth sequence (cycles 376 through 400).

"e. Stabilize the battery temperature at 24°C (75°F) and discharge at the 1C-rate to 20.0 volts.

"f. Constant potential charge the battery in accordance with 4.4.2.

"g. Discharge the battery at the 1C-rate to 20.0 volts.

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"h. Examine the battery for the requirements of 3.8.9.1.

"4.5.10.2 Conformance inspection. Constant potential charge the battery in accordance with 4.4.2 at room temperature. Subject the battery to 100 cycles of discharge-charge-open circuit (rest) at ambient condition of 4.4.1. Each cycle shall consist of 1.0-hour, 1.0 C-rate discharge, followed immediately by a 2.0-hour constant potential charge in accordance with 4.4.2, followed by a 1.0-hour rest period. Examine the battery for the requirements of 3.8.9.2."

8.11 Modify the following paragraphs:

4.5.19 Vibration test. Delete the text and insert: "This test consists of the following steps.

"a. Mount the battery, including electrical connector and cables, to the vibration test apparatus in accordance with 4.4.5.

"b. Constant potential charge each battery in accordance with 4.4.2.

"c. Discharge each battery at the 1C-rate for 10.0 minutes. Monitor the battery's terminal voltage during discharge. Examine the battery for the requirements of 3.8.18.

"d. Constant potential charge each battery in accordance with 4.4.2 during the following vibration. Subject battery sample 2 to the random vibration test levels shown on figure 4 of the specification sheet; the vibration test time for each axis shall be 60.0 ± 3.0 minutes. Subject sample 3 to the sinusoidal vibration test levels shown on figure 5 of the specification sheet; the test time for each axis shall be 30.0 ± 1.5 minutes. Conduct the vibration along each of the battery's three mutually perpendicular axes. After each axis of vibration, discharge the battery at the 1C-rate for 10.0 minutes. Monitor the battery's terminal voltage during discharge. Examine the battery for the requirements of 3.8.18.

"e. Dismount the battery from the vibration test apparatus.

"f. Discharge each battery at the 6.5C-rate for 3.0 minutes.

"g. Constant potential charge each battery in accordance with 4.4.2.

"h. Discharge each battery at the 1C-rate to 20.0 volts.

"i. Constant potential charge each battery in accordance with 4.4.2.

"j. Examine the battery for the requirements of 3.8.18."

4.5.25 Shock test (crash safety). Insert the following after the first sentence of step 4.5.25.b: "Resilient mounting shall not be used." Insert the following before the first sentence of step 4.5.25.c: "Discharge the battery at a C/2 rate during the shock test." Delete the last sentence of step 4.5.25.c and insert: "Deliver three shocks in both directions of each orthogonal axis for a total of 15 impacts."

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8.12 Add the following paragraphs:

"4.5.28.1 Aircraft turnaround loads. Subject samples 2 and 3 to the test of 4.5.28. The test temperatures for sample 2 shall be -26°C (-15°F) and 68°C (154°F) for steps 4.5.28.b and 4.5.28.c. The test temperature for sample 3 shall be 24°C (75°F) for steps 4.5.28.b and 4.5.28.c. The discharge of 4.5.28.c shall be in accordance with figure 3 of the specification sheet. Examine the battery for the requirements of 3.8.16.

"4.5.28.2 Aircraft start-up loads. Perform the test of 4.5.28 two times. The test temperature for steps 4.5.28.b and 4.5.28.c shall be -30°C (-22°F) on the first sequence and 68°C (154°F) on the second sequence. The discharge of 4.5.28.c shall be in accordance with figure 6 of the specification sheet. Examine the battery for the requirements of 3.8.16.

"4.5.28.3 Aircraft emergency loads. Subject samples 2 and 3 to the test of 4.5.28. The test temperatures for sample 2 shall be -18°C (0°F) and 50°C (122°F) for steps 4.5.28.b and 4.5.28.c. The test temperature for sample 3 shall be 24°C (75°F) for steps 4.5.28.b and 4.5.28.c. The discharge of 4.5.28.c shall be 71 amperes for 4.0 minutes. Examine the battery for the requirements of 3.8.16."

8.13 Modify the following paragraph:

4.5.29 Cold and hot temperature charge test. Delete the title and insert: "Ultra-cold temperature charge test." The test temperature for 4.5.29.b and 4.5.29.d shall be -40°C (-40°F). The discharges of 4.5.29.c and 4.5.29.f shall be conducted at the 1C-rate to 20.0 volts.

8.14 Add the following paragraphs:

"4.5.32 Hot temperature charge test. This test consists of the following steps.

"a. Constant potential charge each sample in accordance with 4.4.2.

"b. Stabilize the temperature of sample 1 at 50°C (122°F) and sample 2 at 68°C (154°F) (see 4.4.1). Keep the ambient air temperature at the specified values for each sample while discharging each sample at the 1C-rate to 20.0 volts. Determine the capacity delivered by each sample.

"c. Stabilize the temperature of both sample 1 and sample 2 at 24°C (75°F) (see 4.4.1). Keep the ambient air temperature at 24°C (75°F) while constant potential charging each sample in accordance with 4.4.2.

"d. Stabilize the temperature of sample 1 at 50°C (122°F) and sample 2 at 68°C (154°F) (see 4.4.1). Keep the ambient air temperature at the specified values for each sample while discharging each sample in accordance with figure 3 of the specification sheet. Determine the terminal voltage of each sample during discharge.

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"e. Stabilize the temperature of sample 1 at 50°C (122°F) and sample 2 at 68°C (154°F) (see 4.4.1). Keep the ambient air temperature at the specified values for each sample while constant potential charging each sample in accordance with 4.4.2.

"f. Immediately after completing step 4.5.32.e, place each battery in a 24°C (75°F) environment and discharge each battery at the 1C-rate to 20.0 volts.

"g. Examine the battery for the requirements of 3.8.19.

"6.11.7 Additional application information. The following information is provided for users of the D8565/14-1 battery.

"6.11.7.1 Aircraft charging system. The aircraft electrical system used for charging the battery will provide 28.25 ± 0.50 volts into a load of 50.0 ± 1.0 amperes (see requirement 4 of the specification sheet).

"6.11.7.2 Initial installation date. The initial installer will write the date the battery is first installed in an aircraft (date first placed in service) in the space provided on the battery's identification marking (see 3.6.3 and figure 1).

"6.11.7.3 Service life. The battery's service life (useful life) in aircraft is limited to 2 years. A battery will not be used in aircraft after 2 years from the date first installed in an aircraft (see requirement 6 of the specification sheet).

"6.11.7.4 Removal date from aircraft. The battery will be removed from aircraft service 2 years after the date it was first installed in an aircraft.

"6.11.7.5 Uses other than for aircraft. The battery may be used for applications other than aircraft, such as ground applications, beyond the specified 2-year service life limitation if the battery is deemed satisfactory for such purposes by the user activity.

"6.11.7.6 Maintenance-free. The battery will require no periodically scheduled maintenance during its service life (see 6.15.10).

"6.11.8 Environmental characteristics. Battery performance should not be adversely affected when operating in a sand and/or dust environment. Battery materials should not support fungus growth. The battery should operate safely in an atmosphere containing ignitable or explosive fumes."

Add the following new paragraph:

“8.14 Amendment notations. The margins of this specification sheet are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.”

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8.15 Modify the following table:

TABLE III. Capacity performance requirements. In requirement (1), change "18.0 volts" to "20.0 volts".

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TABLE I. Qualification inspection and conformance inspection of batteries.

Test number	Examinations and tests	Qualification inspection					Conformance inspection		
		Sample number			Requirement paragraph	Test paragraph	Group A sample 1 100%	Group B sample number	
		1	2	3				1	2
1	INCOMING INSPECTION	X	X	X	3.8.1	4.5.1		X	X
2	VISUAL AND MECHANICAL EXAMINATION	X	X	X	Reqmt 5, 3.5, 3.6, 3.8.2, and table II [1/]	4.5.2	X	X	X
3	DIMENSIONS AND WEIGHT	X	X	X	Reqmt 2 & 3.8.3	4.5.3	X[2/]	X	X
4	DIELECTRIC STRENGTH AND INSULATION RESISTANCE TEST	X	X	X	3.7 & 3.8.4	4.5.4	X[2/]	X	X
5	STRENGTH OF VENT TUBES TEST	X	X	X	3.7 & 3.8.5	4.5.5	X[2/]	X	X
6	ELECTRICAL TEST PREPARATION	X	X	X	~~3.8.2~~	4.5.6 [3/]		X	X
7	CAPACITY DISCHARGE	X	X	X	Reqmt 3, ~~3.8.2 &~~ 3.8.6 & Reqmt (1) of table III [3/]	4.5.7 [3/]	X	X	X
8	AIRCRAFT TURNAROUND LOADS	~~X~~	X	X	3.8.16	4.5.28.1			
9	AIRCRAFT START-UP LOADS	~~X~~	~~X~~	X	3.8.16	4.5.28.2			
10	AIRCRAFT EMERGENCY LOADS	~~X~~	X	X	3.8.16	4.5.28.3			
11	HOT-TEMPERATURE CHARGE TEST	X	X		3.7 & 3.8.19	4.5.32			
12	ULTRA-COLD TEMPERATURE CHARGE TEST			X	3.7 & 3.8.17	4.5.29			
13	CYCLING TEST	X			(Qual) 3.8.9.1 & Reqmt 6	4.5.10.1			
					(Conformance) Reqmt 4, 3.8.9.2 & Reqmt (1) of table III [3/]	4.5.10.2		X	
14	DISCHARGE WHILE INVERTED		X		3.5.9 & Reqmt (5) of table III	4.5.13			
15	ALTITUDE TEST		X		3.5.9, 3.7, 3.8.10	4.5.14			X
16	SHOCK TEST (BASIC DESIGN)	X			3.7 & 3.8.10	4.5.15		X	

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TABLE I. Qualification inspection and conformance inspection of batteries - Continued.

Test number	Examinations and tests	Qualification inspection					Conformance inspection		
		Sample number			Requirement paragraph	Test paragraph	Group A sample 1 100%	Group B sample number	
		1	2	3				1	2
17	TEMPERATURE SHOCK TEST	X			3.7 & 3.8.10	4.5.16		X	
18	TEMPERATURE RISE AND FLOAT TEST		X		3.7, 3.8.11, & Reqmt (1) of table III [3/]	4.5.17 [3/]			X
19	BATTERY GAS EMISSION TEST		X		3.8.10 & 3.8.12	4.5.18			
20	GROUND STORAGE TEST			X [4/]	3.8.13.1	4.5.22 [3/]			
21	VIBRATION TEST		X	X	3.7, 3.8.10 & 3.8.18	4.5.19			
22	HUMIDITY TEST		X		3.7 & 3.8.10	4.5.20			X
23	SALT FOG TEST	X			3.7 & 3.8.10	4.5.21		X	
24	DEEP DISCHARGE RECOVERY TEST			X	3.7 & 3.8.13.3	4.5.27			
25	PHYSICAL INTEGRITY AT HIGH TEMPERATURE TEST	X			3.7, 3.8.10 & Reqmt (1) of table III [3/]	4.5.23			
26	SHOCK TEST (CRASH SAFETY)	X			3.8.10 & 3.8.15	4.5.25			
26	IMPACT RESISTANCE TEST (NONMETALLIC BATTERY CONTAINER)		[5/]		3.8.14	4.5.24			
27	SHELF LIFE TEST			X	3.8.9 & 3.8.13.2	4.5.26			
28	FINAL EXAMINATION	X	X	X	3.7 & 3.8.4	4.5.31		X	X

[1/] Table II or III of MIL-PRF-8565.
[2/] See sample information in test paragraph.
[3/] As modified by Requirement 8.2 of MIL-PRF-8565/14.
[4/] The manufacturer requesting qualification to this specification may submit an additional (fourth) test sample to expedite the qualification test program.
[5/] For nonmetallic battery container designs, two outer nonmetallic battery containers shall be furnished for inspection.

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TABLE II. Life cycling temperature conditions (qualification inspection).

Cycle Number	Temperature °C (°F)
1 - 25	24 (75)
26 - 50	43 (110)
51 - 75	-18 (0)
76 - 100	24 (75)
101 - 125	43 (110)
126 - 150	-18 (0)
151 - 175	24 (75)
176 - 200	43 (110)
201 - 225	-18 (0)
226 - 250	24 (75)
251 - 275	43 (110)
276 - 300	-18 (0)
301 - 325	24 (75)
326 - 350	43 (110)
351 - 375	-18 (0)
376 - 400	24 (75)

Custodian:
Navy - AS

Preparing activity:
Navy - AS

(Project 6140-2006-002)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:34Z
Check the source to verify that this is the current version before use.